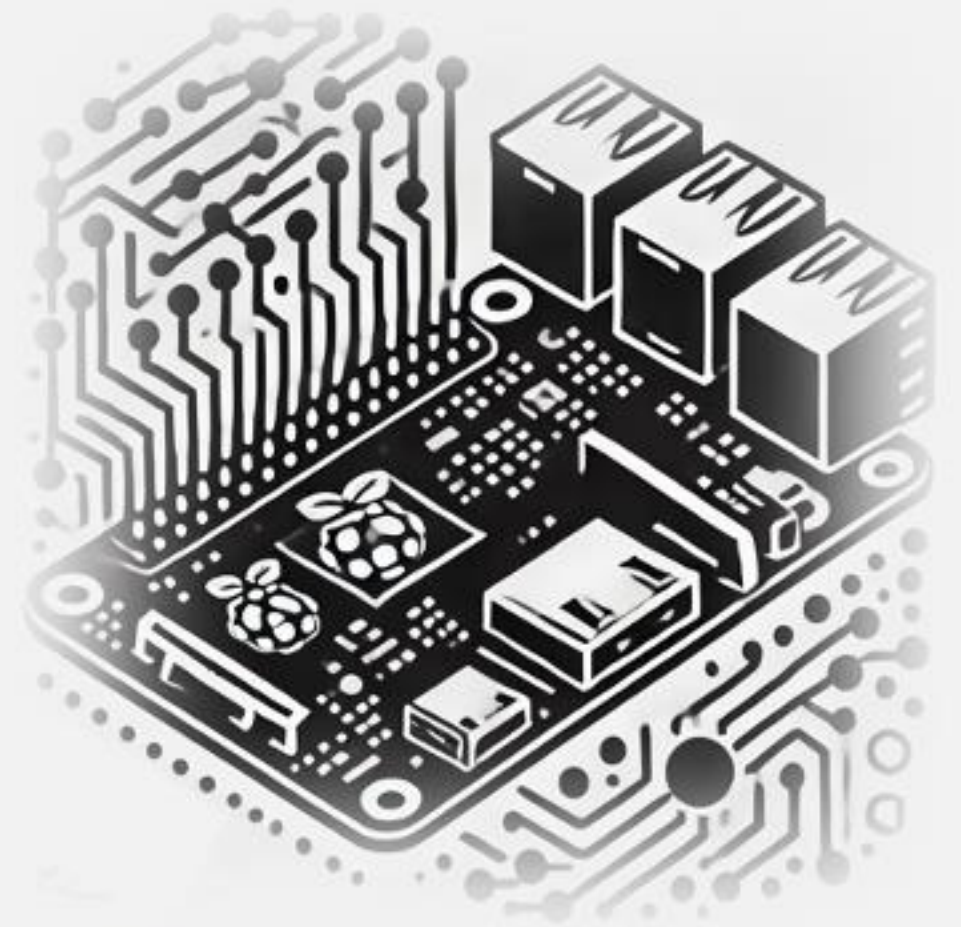


# IESTI05 – Edge AI

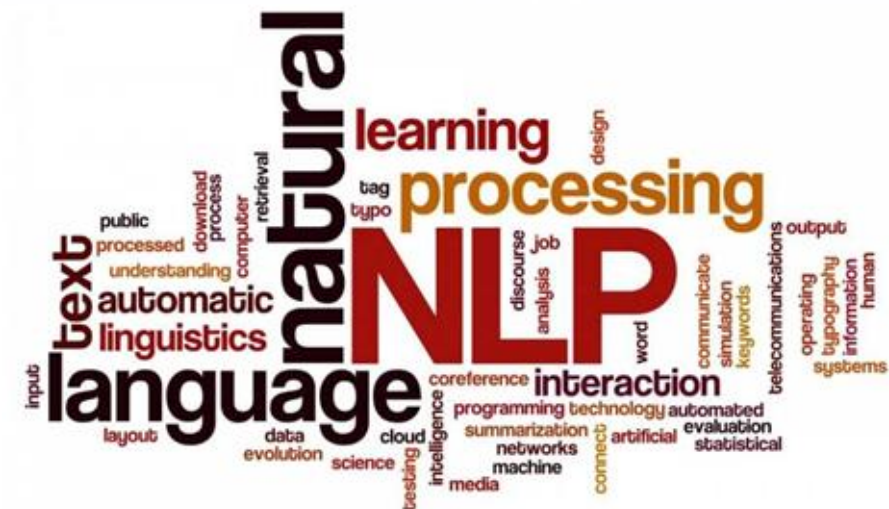
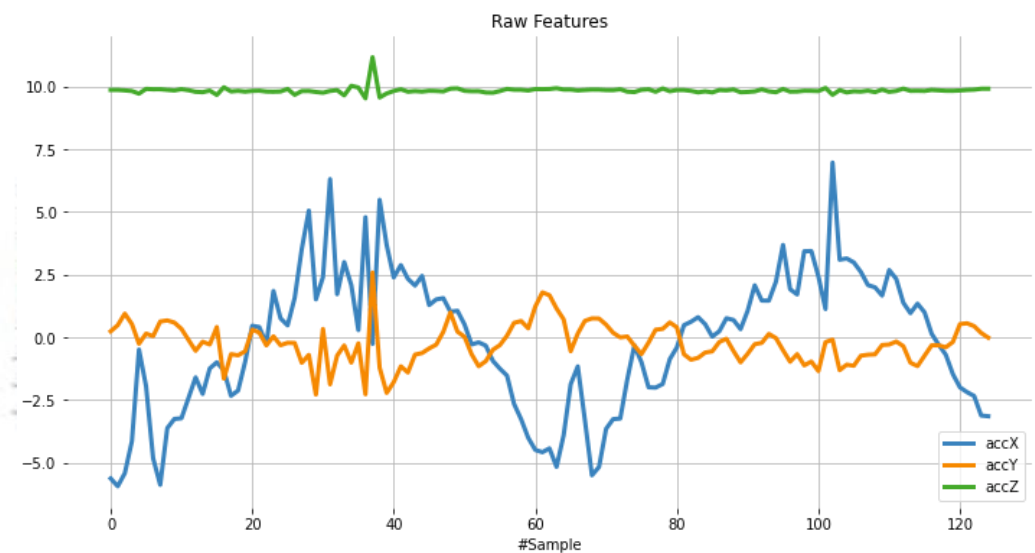
## Machine Learning System Engineering

---

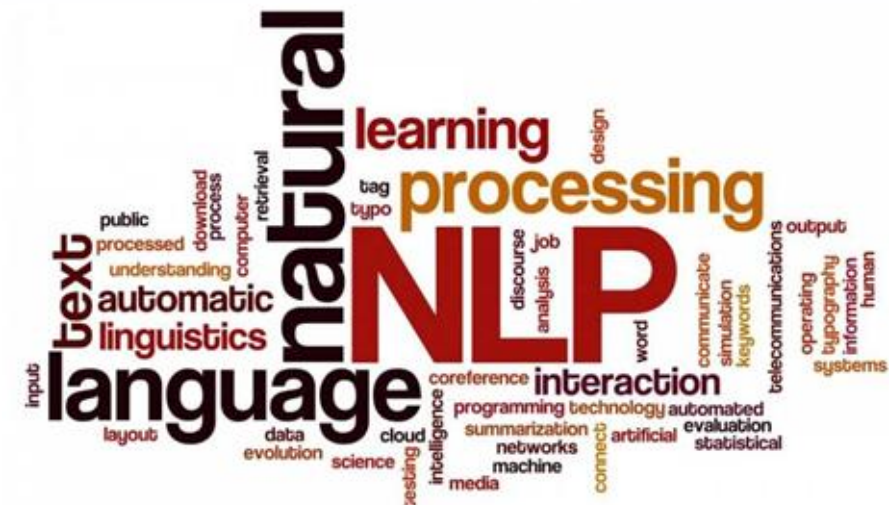
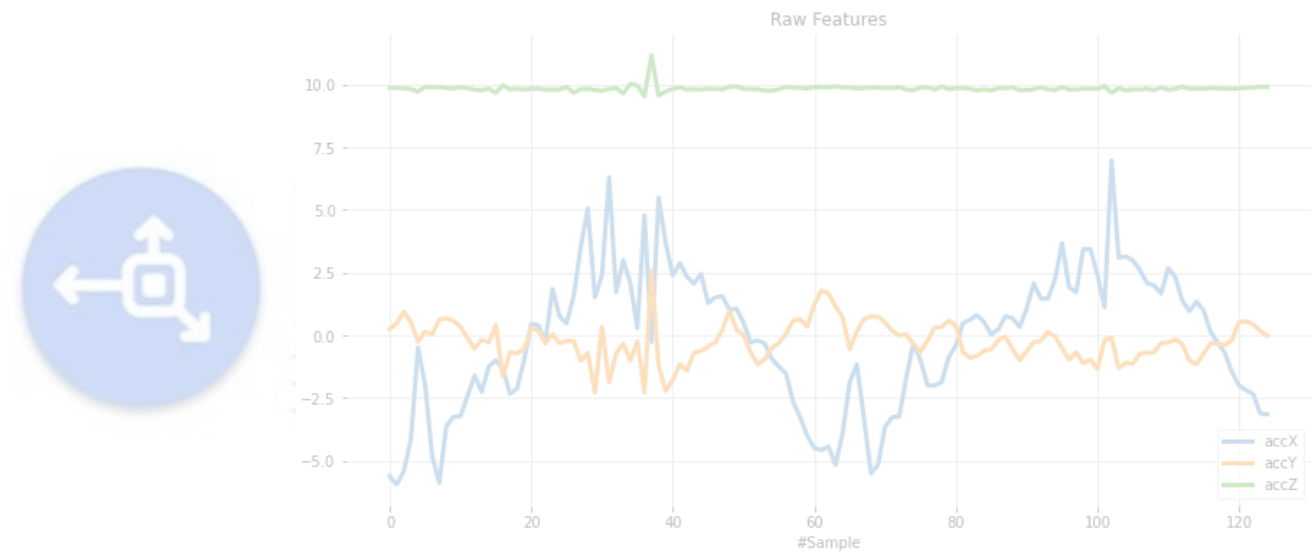
### 15. Text Generation – RNN The VerneBot



# Unstructured Data



# Unstructured Data - Text





# Neural Network Architectures

Vibration  
Analysis



Image  
Classification



Text  
Generation



Image  
Generation



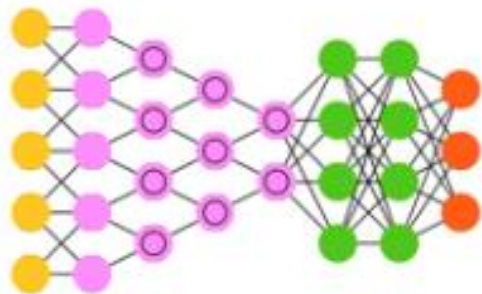
Large Language  
Models- LLMs



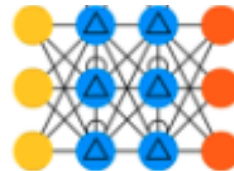
**MLP** - Deep Neural Network



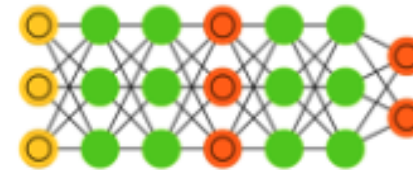
**CNN** - Convolutional NN



**RNN** - Recurrent NN (GRU/LSTM)



**GAN** - Generative Adversarial N.



**AN** - Attention (Transformers)





# Neural Network Architectures

Vibration  
Analysis



Image  
Classification



Text  
Generation



Image  
Generation



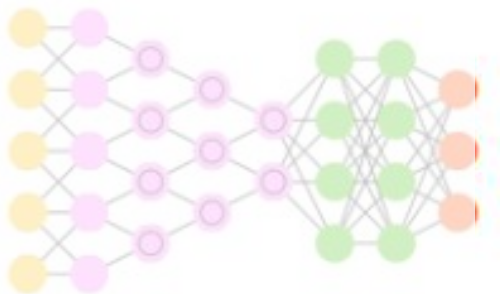
Large Language  
Models- LLMs



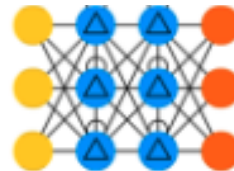
**MLP** - Deep Neural Network



**CNN** - Convolutional NN



**RNN** - Recurrent NN (GRU/LSTM)

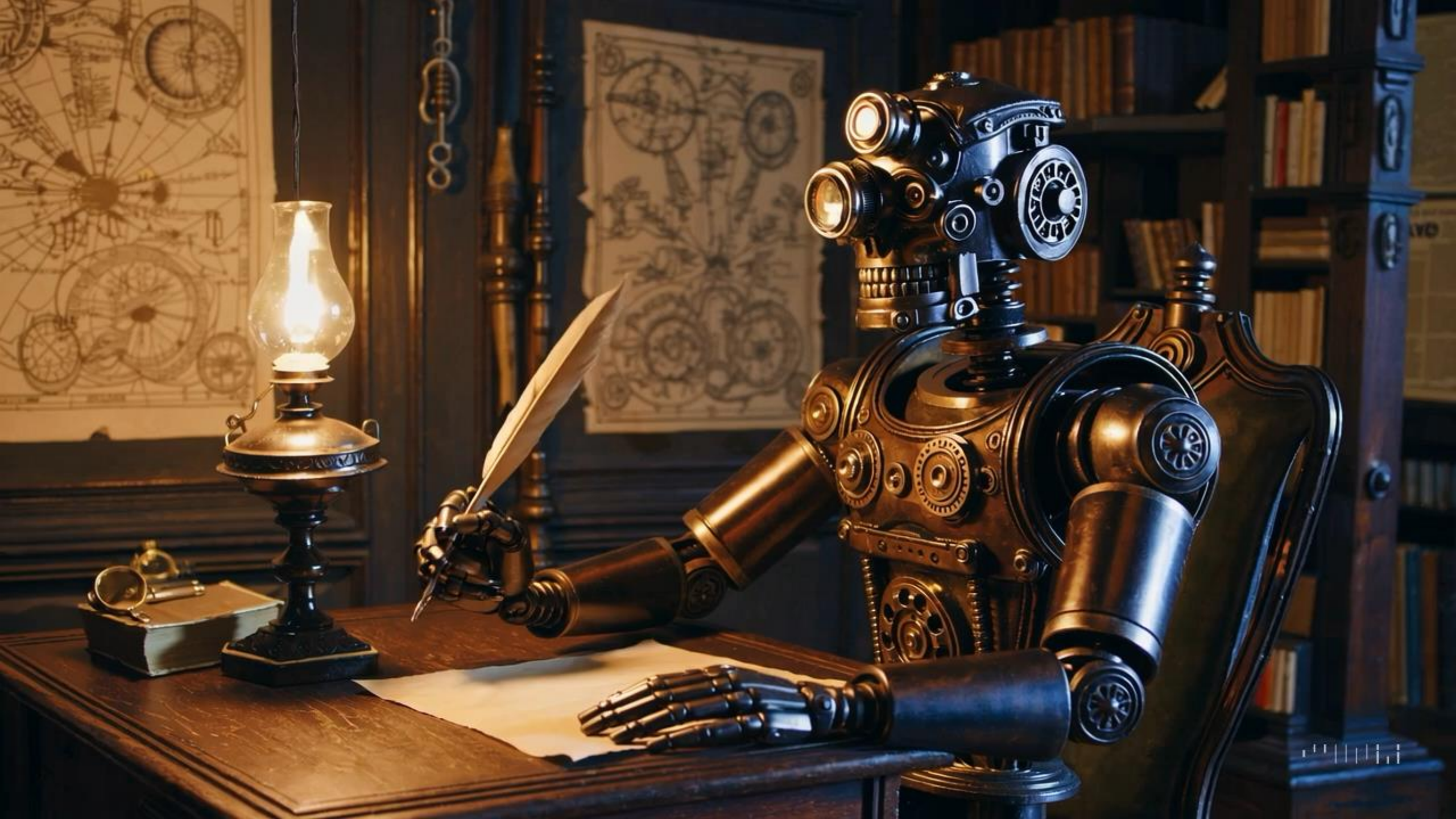


**GAN** - Generative Adversarial N.



**AN** - Attention (Transformers)





20\_JulesVerneBot\_Generatin x +

← → ↺ 🏠

https://colab.research.google.com/github/Mjrovai/TinyML4D/blob/main/WALC\_2024/notebooks/20\_JulesVerneBot\_Generating\_English\_Texts\_with...

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🔍 Commands + Code + Text

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🖼️

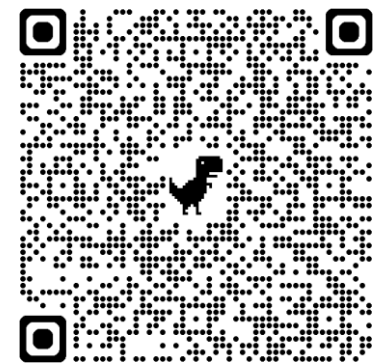
⌵ Generating Text with RNNs: The Jules Verne Bot

- By Marcelo Rovai @Nov24
- With code support from **Claude 3.5 Sonnet** and **ChatGPT 4o**



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Project Gutenberg

# DATASET

Total characters:

- 5,768,791

Unique characters:

- 123



1. 'A Journey to the Centre of the Earth'
2. 'In Search of the Castaways'
3. 'An Antarctic Mystery'
4. 'In the year 2889'
5. 'Around the World in Eighty Days'
6. 'Michael Strogoff'
7. 'Five Weeks in a Balloon'
8. 'The Mysterious Island'
9. 'From the Earth to the Moon'
10. 'Twenty Thousand Leagues under the Sea'

# TOKENIZATION AND VOCABULARY

Conversion of text  
into numeric tokens.



Character-level  
tokenization: 123  
unique characters.



Example: “The Project” →  
[122 52 69 66 1 48 79 76 71 66 64 81].



# TRAINING SEQUENCES



Goal: Predict the next character in a sequence.



Length of the sequence: 120 characters (~ paragraph).



Input 'Hello my nam'

Output 'ello my name'.

# EMBEDDING

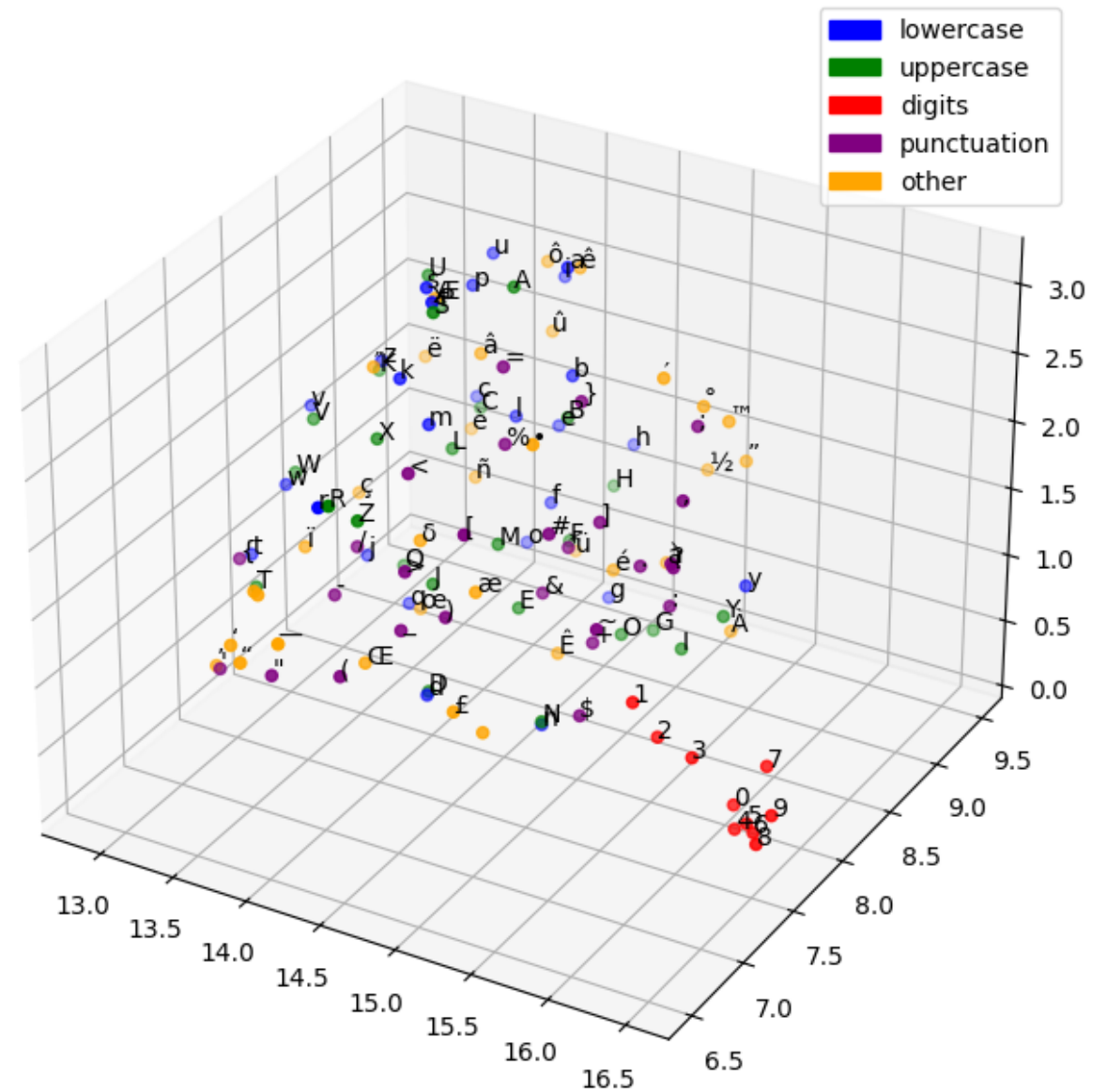
Each character is represented as a vector of 256 dimensions.



*Embedding* captures relationships between characters in dense vectors.

# EMBEDDING

3D Projection of Character Embeddings Using UMAP



# MODEL ARCHITECTURE

Embedding Layer:  
Converts characters into  
dense vectors.

A light blue downward-pointing arrow indicating the flow from the Embedding Layer to the RNN/GRU Layer.

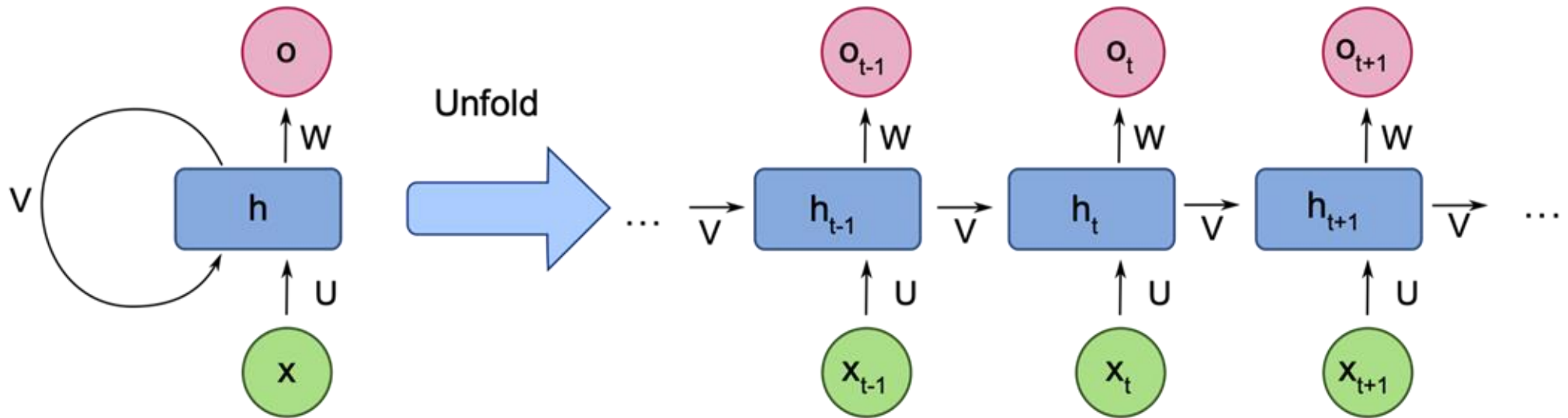
RNN/GRU Layer (1024  
units): Learn from  
sequences.

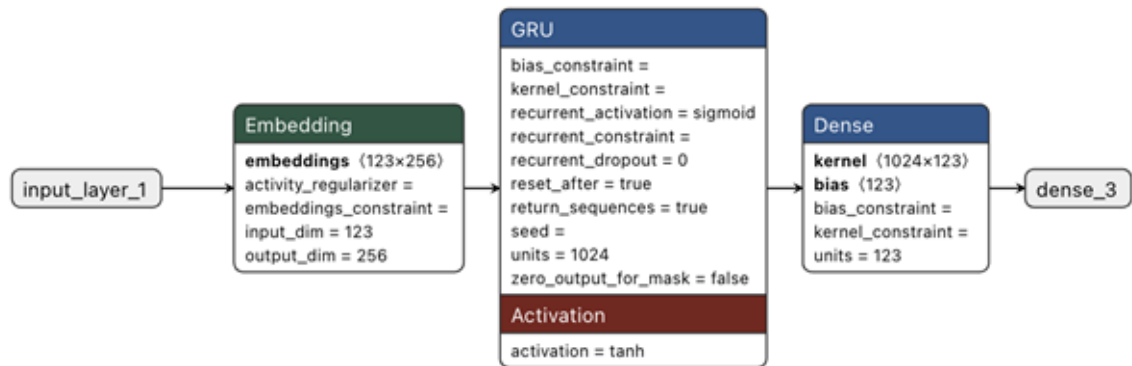
A light blue downward-pointing arrow indicating the flow from the RNN/GRU Layer to the Dense layer.

Dense layer: Generates  
probabilities for each  
character (123).

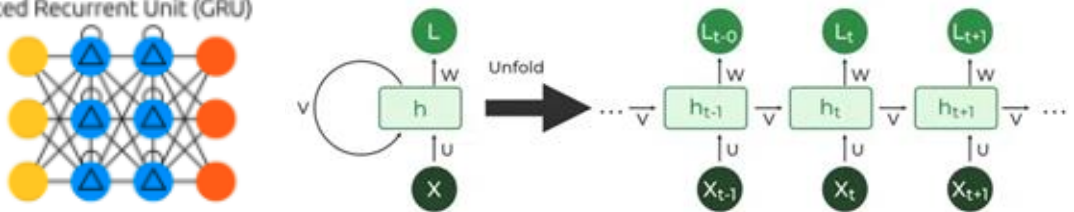
# Deep Learning models (or artificial neural networks)

**Recurrent Neural Networks (RNNs):** Designed for **sequential data like time series or text**, these networks use their internal state (memory) to process sequences of inputs.





Gated Recurrent Unit (GRU)



Model: "sequential\_4"

| Layer (type)                              | Output Shape   | Param #   |
|-------------------------------------------|----------------|-----------|
| embedding_4 ( <a href="#">Embedding</a> ) | (1, 120, 256)  | 31,488    |
| gru_3 ( <a href="#">GRU</a> )             | (1, 120, 1024) | 3,938,304 |
| dense_3 ( <a href="#">Dense</a> )         | (1, 120, 123)  | 126,075   |

Total params: 4,095,867 (15.62 MB)  
 Trainable params: 4,095,867 (15.62 MB)  
 Non-trainable params: 0 (0.00 B)

# RNN MODEL (RECURRENT)

# MODEL TRAINING

Loss Function: Categorical  
Sparse Crossentropy

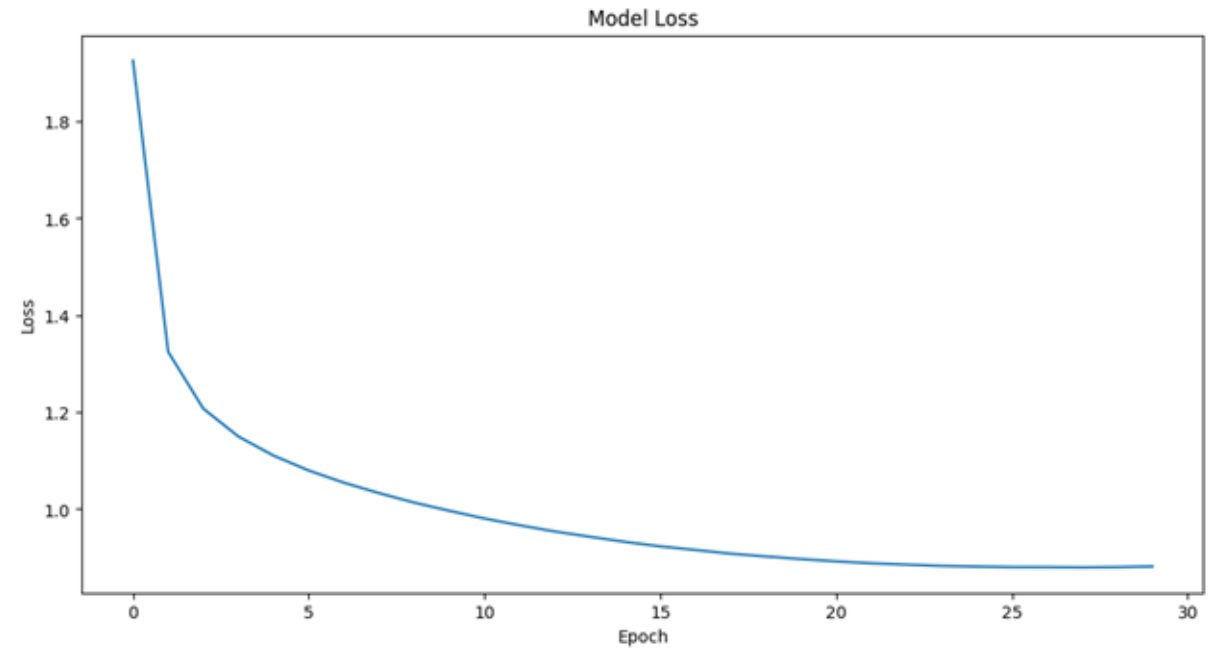
Optimizer: Adam

Epochs: 30

batch size: 128

buffer size: 10,000 (for  
dataset shuffling)

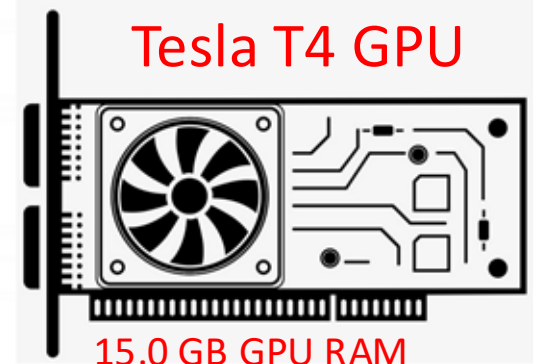
Monitoring loss reduction  
over time.



(33 minutes for training)

Google  
colab

Tesla T4 GPU



15.0 GB GPU RAM

# TEXT GENERATION

The template generates text character by character from an initial text:

"THE FLYING  
SUBMARINE".

Temperature controls randomness (0.5 for predictable, 1.0 for creative text).

Generated text with temperature 0.5:

THE FLYING SUBMARINE

CHAPTER 100 VENTANTILE

This eBook is for the use of anyone anywhere in the United States and most other parts of the earth and miserable eruptions. The solar rays should be entirely under the shock of the intensity of the sea. We were all sorts. Are we to prepare for our feelings?"

"I can never see them a good geographer," said Mary.

"Well, then, John, for I get to the Pampas, that we ought to obey the same time. In the country of this latitude changed my brother, and the \_Nautilus\_ floated in a sea which contained the rudder and lower colour visibly. The loiter was a fatalint region the two scientific discoverers. Several times turning toward the river, the cry of doors and over an inclined plains of the Angara, with a threatening water and disappeared in the midst of the solar rays.

The weather was spread and strewn with closed bottoms which soon appeared that the unexpected sheets of wind was soon and linen, and the whole seas were again landed on the subject of the natives, and the prisoners were successively assuming the sides of this agreement for fifteen days with a threatening voice.

The clouds had disappeared to the ground, and the river and the ship's conditions of the ship's course was still standing, but in his daring explorers, and the sailor thought for the sudden discovery.

"There are no trees, and a half an hour or the sun will soon be carried off they were bringing a special track."

"As you see, my dear Helena, who was the matter of despair?" cried the captain.

"The raft we had done now, captain, and I have a reasonable face of the shipwrecked creek, had been discovered at the entrance of the world."

# CHALLENGES AND LIMITATIONS

*Limited **context window** (120 characters).*

Difficulty in maintaining coherence in long texts.

**Character-level modeling** vs. word-level modeling.



# Neural Network Architectures

Vibration  
Analysis



Image  
Classification



Text  
Generation



Image  
Generation



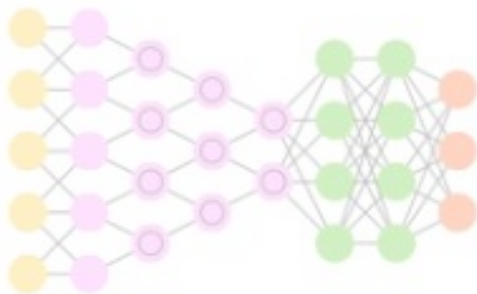
Large Language  
Models- LLMs



**MLP** - Deep Neural Network



**CNN** - Convolutional NN



**RNN** - Recurrent NN (GRU/LSTM)



**GAN** - Generative Adversarial N.



**AN** - Attention (Transformers)



# THE TRANSFORMER EXPERIMENT

*Same books, same 123-  
character vocabulary, same size  
(2% apart).*

Only the architecture changes:  
recurrence against attention.

**Validation:** 5% held out from the  
middle of every novel.

## RESULTS ON HELD-OUT TEXT

*At the same window: **RNN**  
**1.146, Transformer 1.102.***

With a 256-character window,  
1.034 — context bought more.

**Three seeds each:** the gap is  
~22x the seed-to-seed spread.

# WHAT THE KV CACHE TAUGHT US

*A **KV cache** makes each character cost one pass, not a replay.*

With learned positions it cannot wrap, or the window reads rotated.

**Rebuilt every 64 characters:** real words went from 54% to 97%.

TRY IT IN THE  
BROWSER

*[mjrovai.github.io/Jules\\_Verne](https://mjrovai.github.io/Jules_Verne)  
— three models, no server.*

“Run all three” writes one seed  
through each of them.

**Code and notebooks:**

[https://github.com/Mjrovai/Jules\\_Verne](https://github.com/Mjrovai/Jules_Verne)





# CONNECTING WITH MODERN LANGUAGE MODELS



## Our Model (VerneBot) :

- Training data: **5.8 million characters (bytes)** (10 books).
- **4 million parameters**,
- Character-level tokenization (**123**)
- **RNN** architecture.

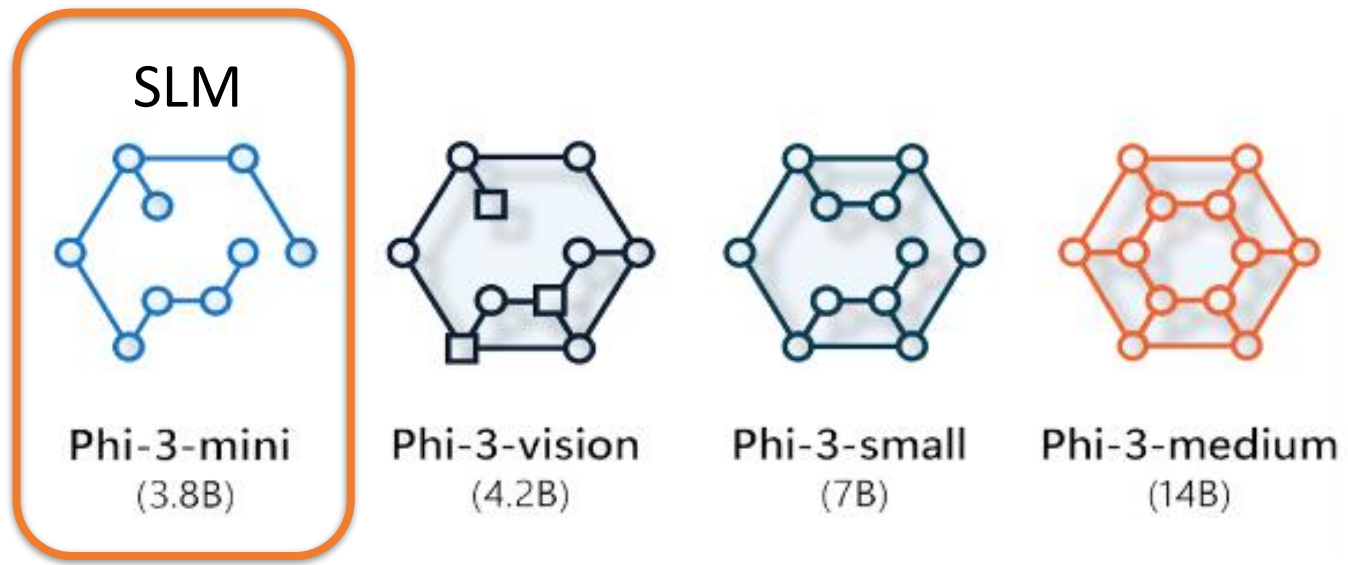


## Open AI GPT-3 (2020):

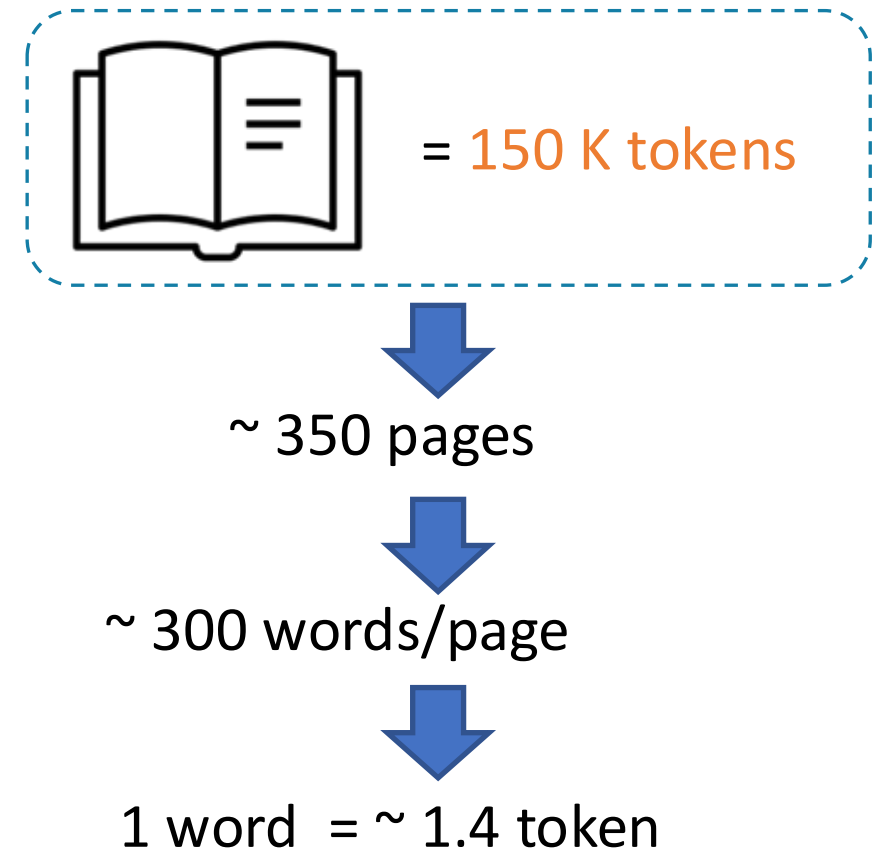
- Training data: **~570 GB of filtered text**
- **175 billion parameters**,
- Subword tokenization (**50,257 tokens**),
- **Transformer**, 2,048-token context.



Modern models handle long-range dependencies better.



- **Architecture:** Transformer – 3.8 Billion Parameters  
**Inputs:** Text.  
**Context length:** 128k tokens  
**GPU:** 512 H100-80G  
**Training time:** 7 days  
**Training data:** 3.3 Trillion tokens\*\*



\*\* Equivalent to 23 million books, that is:  
17% of all the books in the world

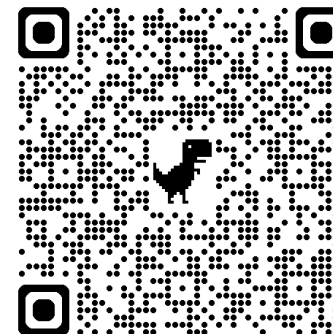
| Aspect           | Jules Verne Bot | GPT-3 (2020)      | Phi-3-mini (2024) |
|------------------|-----------------|-------------------|-------------------|
| Architecture     | RNN (GRU)       | Transformer       | Transformer       |
| Parameters       | 4 million       | 175 billion       | 3.8 billion       |
| Training Data    | 5.8M characters | 45TB text         | 3.3T tokens       |
| Context Length   | 120 characters  | 2,048 tokens      | 128,000 tokens    |
| Tokenization     | Character-level | Subword (BPE)     | Subword           |
| Training Time    | 33 minutes      | Months            | 7 days            |
| GPU Requirements | 1 Tesla T4      | Thousands of GPUs | 512 H100 GPUs     |

## Scale Comparison

Preview

Code

Blame



# Machado de Assis Bot

Portuguese Text Generation with Artificial Neural Networks



- Project developed by Marcelo Rovai, based on lessons learned with the great teacher [José Portilla](#) during his fantastic course: [Complete Tensorflow 2 and Keras Deep Learning Bootcamp](#)
- The Artificial network developed on this project, generates text, character by character. Please see this following site, for a great general visual explanation: [The Unreasonable Effectiveness of Recurrent Neural Networks](#) and also check the [TF site for details](#).

# Questions?

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**UNIFEI**